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The mode your accelerometer cannot see



The SLS core stage and modal-test hardware inside the Vehicle Assembly Building in August 2021. Photo: NASA, KSC-20210813-PH-JBS01_0015. This image shows the test configuration; it does not identify individual sensor locations. NASA image source [1].

The rocket is full size. The measurement is not.

NASA used integrated modal testing before Artemis I to measure the Space Launch System natural frequencies, flexes and vibration patterns needed by flight software and navigation models [1].

Even on a structure this large, the test sees only the coordinates selected by the instrumentation plan. The unmeasured structure must be inferred from those samples.

A mode can be present and still be absent

A modal survey is sometimes described as if the structure simply reveals its modes when it is shaken. The reality is more selective. A sensor measures one physical coordinate, along one axis, at one location. A shaker or hammer injects force at another coordinate. If either point lies near a node of a target mode, that mode may contribute almost nothing to the measured frequency-response function.

Today’s objective is to make that selection effect tangible: distinguish the full mode shape from its measured projection, show how sparse spatial sampling can make two different modes look identical, and turn the result into better pretest analysis, correlation and model-validation decisions.

A sensor measures a coordinate, not a mode

Consider the familiar transverse modes of an ideal simply supported beam. With x measured along span L, the r-th normalized shape is

$$\phi_r(x) = \sin\left(\frac{r\pi x}{L}\right)$$

The complete functions are orthogonal over the continuous span. But an accelerometer array does not observe the continuous span. It evaluates the shapes only at selected coordinates and directions. If S is the Boolean or interpolation matrix that maps model degrees of freedom into measured channels, the experimental view of mode r is

$$\widetilde{\boldsymbol{\phi}}_r = \mathbf{S}\boldsymbol{\phi}_r$$

Everything downstream sees this projected vector. A local node makes one component small. A sparse or symmetric array can do something subtler: it can project two distinct full-field shapes into proportional vectors. That is spatial aliasing.

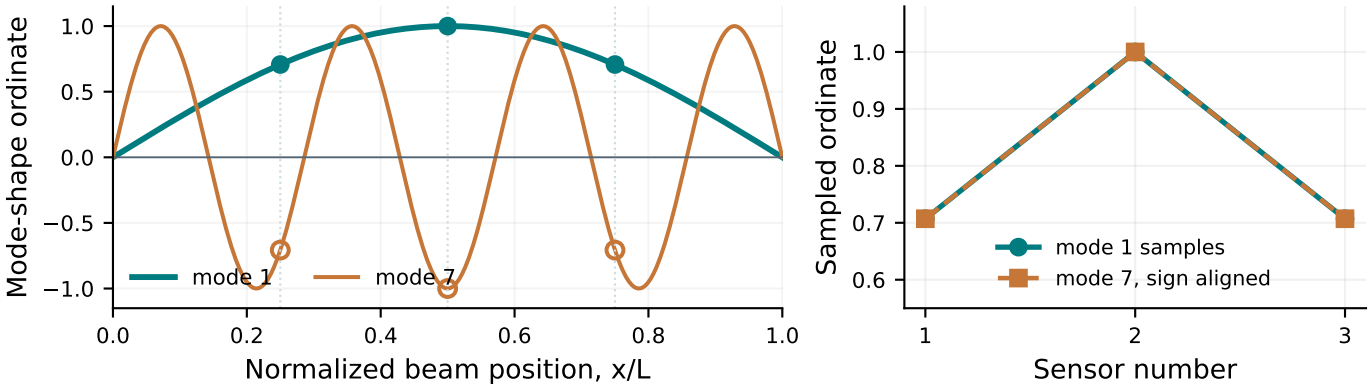


Figure 1. Original analytical illustration. Modes 1 and 7 of an ideal simply supported beam are different continuous shapes, but at x/L = 0.25, 0.50 and 0.75 their sampled vectors are exact negatives. Because a mode-shape sign is arbitrary, the measured shapes are indistinguishable at those coordinates.

This is not an argument against evenly spaced sensors in every test. It is an argument against choosing a pattern without checking the target modes. Symmetry that looks tidy on a drawing can preserve the very ambiguity the test is meant to remove.

The same logic applies to direction. An accelerometer placed at an antinode but oriented nearly orthogonal to the local modal motion can be almost as blind as one placed on a geometric node. Rotational content, local panel motion and internal hardware modes may also be weakly represented by an array designed only around global translation.

A perfect MAC can be perfectly misleading

The Modal Assurance Criterion compares two shape vectors without caring about scale or sign. For real vectors a and b,

$$\mathrm{MAC}(\mathbf{a},\mathbf{b}) = \frac{|\mathbf{a}^{\mathsf{T}}\mathbf{b}|^2}{(\mathbf{a}^{\mathsf{T}}\mathbf{a})(\mathbf{b}^{\mathsf{T}}\mathbf{b})}$$

MAC equals one when the sampled vectors are proportional and zero when they are orthogonal. That definition is useful, but it says nothing about the coordinates that were omitted. For the three-sensor beam example,

$$\widetilde{\boldsymbol{\phi}}_1 = [0.707,\ 1,\ 0.707]^{\mathsf{T}}, \qquad \widetilde{\boldsymbol{\phi}}_7 = -\widetilde{\boldsymbol{\phi}}_1$$

The sampled MAC is therefore 1.000 even though the continuous shapes are orthogonal. The arithmetic is correct. The inference would be wrong if we claimed that high sampled MAC proved full-field equivalence.

Measurement coordinates, x/L	Channels	Sampled MAC: mode 1 vs 7	Interpretation
0.25, 0.50, 0.75	3	1.000	Perfect alias after sign alignment
0.10, 0.25, 0.50, 0.75	4	0.551	One asymmetric coordinate breaks the identity
0.10, 0.25, 0.42, 0.63, 0.81	5	0.014	Irregular set strongly separates these two targets

Table 1. Illustrative calculations for the ideal beam. The irregular pattern is not an optimized universal layout; it only demonstrates that information content depends on placement, not channel count alone.

Frequency information often helps separate modes that shape sampling confuses. Mode 1 and mode 7 of an ideal beam are far apart. The risk grows when candidate modes are closely spaced, local modes fall near system modes, or model uncertainty reorders a cluster. In those cases, a high MAC on a sparse common set can hide poor independence rather than demonstrate good correlation.

This is why an auto-MAC or cross-orthogonality matrix should be reviewed before the test, using the proposed measurement set and all target analytical modes. Off-diagonal terms expose modes that the array cannot distinguish. A candidate layout should also be perturbed for plausible model error; a sensor that is excellent only because the pretest node is predicted perfectly is fragile.

The mode must be excited and observed

Sensor placement is only half the experiment. Near a lightly damped mode r, one term of the receptance between response coordinate i and force coordinate j is proportional to the product of two mode-shape ordinates:

$$H_{ij}(\omega) \approx \frac{\phi_{ir}\phi_{jr}}{\omega_r^2 - \omega^2 + 2\,\mathrm{i}\,\zeta_r\omega_r\omega}$$

If either ordinate is near zero, the modal residue is small. Moving the response sensor may help observability; moving the input may help controllability. Multiple inputs and multiple references protect against one unlucky node, but only if their spatial patterns add independent information.

A NASA mirror test shows the logic in practice. A 1.5 m lightweight mirror was suspended on bungees for an approximate free-free condition. Three uniaxial accelerometers were fixed at selected locations while an instrumented hammer roved over 22 points on the mirror. The team identified seven modes from about 414 to 877 Hz and compared measured and predicted shapes with MAC [2]. The arrangement traded many fixed response channels for many input locations; it did not make sensor placement irrelevant.

The published study reported four of seven matched pairs at or above its MAC target of 0.90, one just below and two notably lower [2]. That 0.90 value was a project comparison target, not a universal law. The authors also documented model-form concerns associated with deformed ribs, which is a useful reminder: a low MAC may reflect inadequate spatial sampling, a wrong pairing, measurement error, or a genuinely deficient model. The metric does not diagnose the cause by itself.

At the other end of scale, the International Space Station Node modal survey used more than 400 triaxial accelerometers and over 1,200 channels because internal components as well as the shell had to be represented. NASA’s placement study compared Guyan reduction, kinetic-energy and other selection approaches, and found that methods behaved differently for that constrained configuration [3]. More sensors expanded coverage, but disciplined pretest reduction and cross-orthogonality checks were still required.

The transfer to vehicle safety is direct. A correlated test-analysis model is later asked to predict loads at uninstrumented locations and under flight boundary conditions. If the test array could not distinguish important target modes, numerical agreement at the retained channels may overstate what was actually validated.

Treat instrumentation as part of the model

Start pretest analysis with the decision the modal model must support. Define target modes from loads, controls, pointing, fatigue or clearance needs. Map them through the proposed sensor directions, then review projected auto-MAC, rank, conditioning and expected signal level across plausible model variations.

Do the same on the input side. Compare residues or controllability for candidate shaker and hammer locations, and avoid relying on one input to reach every mode. Include the real fixture, suspension, cables and offload system; a clean free-free model is not the article hanging in the lab.

During test, challenge the data with reciprocity, coherence, repeats and reference consistency. If a target mode is absent, first ask whether the force reached it, the references could see it, and the frequency resolution separated it before declaring the model wrong.

NASA-HDBK-7008 treats modal testing as one element of spacecraft dynamic testing [4]. It is guidance, not a universal requirement; the governing contract and approved test plan control. The broader principle remains: validation evidence is limited to the modes and coordinates the test can observe.

Three takeaways

1. A mode shape measured at sparse coordinates is a projection, not the full structural field.
2. High MAC proves proportionality only on the compared coordinates; spatial aliasing can make different modes look identical.
3. Design sensor and input locations together, against target modes and credible model uncertainty.

Reflection: A predicted mode has no clear resonance in the test FRFs. What is the first interpretation?
Answer: Not “the model invented a mode.” First check whether the excitation and response coordinates give that mode enough residue to be observable, then examine resolution, pairing and model error.

Sources, scope and figure credits

[1] NASA, “SLS Modal Testing: Finding the Range of a Rocket’s Reflexes,” September 24, 2021. Opening photograph: NASA, KSC-20210813-PH-JBS01_0015.

[2] J. B. Knight et al., “Advanced Mirror Technology Development II Modal Test of a 1.5 m Glass Slumped Mirror,” NASA Marshall Space Flight Center, 2018, NASA NTRS 20180006365.

[3] M. L. Tinker, “Accelerometer Placement for the International Space Station Node Modal Test,” AIAA-98-2078, 1998, NASA NTRS 19980111020.

[4] NASA Handbook for Spacecraft Structural Dynamics Testing, 2007, NASA NTRS 20070021678; precursor and technical basis for NASA-HDBK-7008 guidance.

[5] D. C. Kammer, “Sensor Placement for On-Orbit Modal Identification and Correlation of Large Space Structures,” Journal of Guidance, Control, and Dynamics, 14(2), 251-259, 1991. DOI 10.2514/3.20635.